

Random Vectors

Clément Lalanne

What this lecture does

Define four modes of convergence for random vectors.

Characterise convergence in distribution without cumulative distribution functions: the portmanteau theorem replaces them by test functions and by sets, which is what every later proof actually uses.

Turn it into an analytic condition. Characteristic functions convert convergence of laws into pointwise convergence of functions on $\mathbb{R}^d$.

Assemble a calculus. Slutsky, continuous mapping, and the symbols $o_{\mathbb{P}}$ and $O_{\mathbb{P}}$.

Notation for this lecture

The space. All random vectors live on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and take values in $\mathbb{R}^d$ with the Euclidean inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. Measures are capital blackboard letters: $\mathbb{P}$ on Ω , $\mathbb{P}_X$ for the law of X on $\mathbb{R}^d$, $\mathbb{P}^{\otimes n}$ for a product. Densities are lowercase, p, p_θ .

Subscripts. Every expectation, variance and supremum carries what it is taken over: $\mathbb{E}_X, \text{Var}_X, \mathbb{E}_{X_n}$, and $\mathbb{E}_S$ for a sample $S = (X_1, \dots, X_n) \sim \mathbb{P}_X^{\otimes n}$.

Arrows. $X_n \xrightarrow{d} X, X_n \xrightarrow{\mathbb{P}} X, X_n \xrightarrow{a.s.} X, X_n \xrightarrow{L^p} X$, all as $n \rightarrow \infty$.

Sets. For $A \subset \mathbb{R}^d$, $\bar{A}$ is the closure, $\mathring{A}$ the interior, $\partial A = \bar{A} \setminus \mathring{A}$ the boundary, A^c the complement. $B(c, r) = \{x : \|x - c\| < r\}$.

Part I. Four modes of convergence

The cumulative distribution function

Definition. For a random vector $X = (X_1, \dots, X_d)$ of $\mathbb{R}^d$, its cumulative distribution function is

$$F_X : \mathbb{R}^d \rightarrow [0, 1], \quad (x_1, \dots, x_d) \mapsto \mathbb{P}(X_1 \leq x_1, \dots, X_d \leq x_d).$$

Proposition. F_X determines $\mathbb{P}_X$.

Proof. The family $\mathcal{C} = \{(-\infty, x] : x \in \mathbb{R}^d\}$, where $(-\infty, x] = \prod_{j \leq d} (-\infty, x_j]$, is stable under finite intersections and generates $\mathcal{B}(\mathbb{R}^d)$. Two laws agreeing on a generating π -system agree on the σ -field it generates, by the Dynkin lemma.

Four modes of convergence

Definition. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random vectors of $\mathbb{R}^d$ and X a random vector of $\mathbb{R}^d$. We say that X_n converges to X

in distribution, written $X_n \xrightarrow{d} X$, if $F_{X_n}(x) \rightarrow F_X(x)$ for every x at which F_X is continuous.

in probability, written $X_n \xrightarrow{\mathbb{P}} X$, if $\mathbb{P}(\|X_n - X\| \geq \epsilon) \rightarrow 0$ for every $\epsilon > 0$.

almost surely, written $X_n \xrightarrow{a.s.} X$, if $\mathbb{P}(\|X_n - X\| \rightarrow 0) = 1$.

in L^p , $p \geq 1$, written $X_n \xrightarrow{L^p} X$, if $\mathbb{E}_{X_n, X}[\|X_n - X\|^p] \rightarrow 0$.

What each definition needs

Only convergence in distribution is a statement about laws. It involves $\mathbb{P}_{X_n}$ and $\mathbb{P}_X$ and nothing else, so the X_n need not be defined on a common probability space, or on any common space with X . The other three use $X_n - X$, hence require all of them on one $(\Omega, \mathcal{A}, \mathbb{P})$.

Consequence. The limit in distribution can be a law $\mathcal{L}$ on $\mathbb{R}^d$ rather than a random vector, and we write $X_n \xrightarrow{d} \mathcal{L}$. Every statement below holds with that reading, with $\mathbb{P}(X \in A)$ replaced by $\mathcal{L}(A)$.

A limit in distribution is not unique as a random vector, only as a law. If $X_n \xrightarrow{d} X$ and $X' \stackrel{d}{=} X$ then $X_n \xrightarrow{d} X'$, whatever the dependence between X' and the X_n .

Why only continuity points

Take $d = 1$, $X_n = 1/n$ deterministic and $X = 0$. Then $X_n \rightarrow X$ in every reasonable sense, yet

$$F_{X_n}(0) = \mathbb{P}(1/n \leq 0) = 0 \not\rightarrow 1 = F_X(0).$$

The point 0 is the unique discontinuity point of F_X . Requiring convergence there would make even a deterministic sequence converging to a point fail to converge in distribution, so the definition excludes those points. At every $x \neq 0$ we do have $F_{X_n}(x) \rightarrow F_X(x)$.

Remark. The set of discontinuity points of F_X is small: for each coordinate j , at most countably many t satisfy $\mathbb{P}(X_j = t) > 0$, and F_X is continuous at every x whose coordinates avoid those countable sets. So the definition still constrains F_{X_n} on a dense set.

Why countably many. Fix j and $m \geq 1$, and let $A_m = \{t : \mathbb{P}(X_j = t) > 1/m\}$. Distinct $t \in A_m$ give disjoint events $\{X_j = t\}$, whose probabilities sum to at most 1, so A_m has fewer than m elements. The atoms of X_j are $\bigcup_{m \geq 1} A_m$, a countable union of finite sets.

Part II. The portmanteau theorem

The portmanteau theorem

Theorem. Let (X_n) and X be random vectors of $\mathbb{R}^d$. The following are equivalent.

- (i) $X_n \xrightarrow{d} X$.
- (ii) $\mathbb{E}_{X_n}[f(X_n)] \rightarrow \mathbb{E}_X[f(X)]$ for every bounded continuous $f : \mathbb{R}^d \rightarrow \mathbb{R}$.
- (iii) $\mathbb{E}_{X_n}[f(X_n)] \rightarrow \mathbb{E}_X[f(X)]$ for every bounded Lipschitz f .
- (iv) $\liminf_n \mathbb{E}_{X_n}[f(X_n)] \geq \mathbb{E}_X[f(X)]$ for every continuous $f \geq 0$.
- (v) $\liminf_n \mathbb{P}(X_n \in O) \geq \mathbb{P}(X \in O)$ for every open O .
- (vi) $\limsup_n \mathbb{P}(X_n \in F) \leq \mathbb{P}(X \in F)$ for every closed F .
- (vii) $\mathbb{P}(X_n \in B) \rightarrow \mathbb{P}(X \in B)$ for every Borel B with $\mathbb{P}(X \in \partial B) = 0$.

Which condition to use

In practice nobody uses (i). Cumulative distribution functions are unusable in $d > 1$ and are generally impractical.

(ii) and (iii) to prove that something converges, because expectations of test functions are what characteristic functions, Slutsky and continuous mapping all produce.

(vi) to bound a probability from above by the limit, which is how tightness and continuous mapping are proved.

(vii) to read off limit probabilities, which is how a confidence interval is justified in Part VI.

Why (iii) is worth isolating. The class of bounded Lipschitz functions is much smaller than the class of bounded continuous ones, so (iii) is the cheapest sufficient condition.

Proof, (i) implies convergence on rectangles

Assume first F_X continuous everywhere. For $a \leq b$ in $\mathbb{R}^d$ and $(a, b] = \prod_{j \leq d} (a_j, b_j]$, inclusion and exclusion over the 2^d corners gives

$$\mathbb{P}(X \in (a, b]) = \sum_{\epsilon \in \{0,1\}^d} (-1)^{|\epsilon|} F_X(c_\epsilon), \quad (c_\epsilon)_j = \epsilon_j a_j + (1 - \epsilon_j) b_j,$$

and the same identity for each X_n . Every corner is a continuity point, so $\mathbb{P}(X_n \in (a, b]) \rightarrow \mathbb{P}(X \in (a, b])$.

For a rectangle with a closed left face,

$$\mathbb{P}(a < X_n \leq b) \leq \mathbb{P}(a \leq X_n \leq b) \leq \mathbb{P}(a - 2^{-m} \mathbf{1} < X_n \leq b),$$

let $n \rightarrow \infty$ on both ends, then $m \rightarrow \infty$, and use continuity of F_X to close the sandwich. The other faces are identical.

Proof, from rectangles to test functions

Fix $\epsilon > 0$ and a compact rectangle I_ϵ with $\mathbb{P}(X \notin I_\epsilon) \leq \epsilon$. It exists because the events $A_M = \{X \in [-M, M]^d\}$ increase along $M \in \mathbb{N}$, and for a nondecreasing sequence of events $\mathbb{P}(A_M) \rightarrow \mathbb{P}(\bigcup_{M \geq 1} A_M) = 1$.

Let f be bounded continuous, hence uniformly continuous on the compact I_ϵ . Cut I_ϵ into disjoint rectangles $R_1, \dots, R_K$ of diameter below the modulus of f at ϵ , so that f varies by at most ϵ on each, and set

$$\hat{f} = \sum_{i \leq K} \mathbf{1}_{R_i} \inf_{R_i} f, \quad \text{so} \quad |\hat{f}(x) - f(x)| \leq \epsilon + \|f\|_\infty \mathbf{1}\{x \notin I_\epsilon\} \text{ for all } x.$$

Proof, the three term bound

Insert $\hat{f}$ twice and use that it is constant on each R_i :

$$\begin{aligned} |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| &\leq \mathbb{E}_{X_n}|f - \hat{f}|(X_n) + |\mathbb{E}_{X_n}[\hat{f}(X_n)] - \mathbb{E}_X[\hat{f}(X)]| + \mathbb{E}_X|\hat{f} - f|(X) \\ &\leq 2\epsilon + \|f\|_\infty(\mathbb{P}(X_n \notin I_\epsilon) + \mathbb{P}(X \notin I_\epsilon)) + \sum_{i \leq K} |\Delta_{i,n}| \inf_{R_i} |f| \end{aligned}$$

with $\Delta_{i,n} = \mathbb{P}(X_n \in R_i) - \mathbb{P}(X \in R_i) \rightarrow 0$. The last term is a finite sum, so it vanishes, and the middle one tends to $2\mathbb{P}(X \notin I_\epsilon) \leq 2\epsilon$. Hence $\limsup_n |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| \leq 2\epsilon(1 + \|f\|_\infty)$, for every ϵ .

The case of a discontinuous limit. If F_X is not continuous everywhere, the argument is unchanged once the rectangles are chosen well. Indeed, F_X only has countably many discontinuity points, and we simply have to make sure to take rectangles in the previous proof with faces that do not contain the previous discontinuity points.

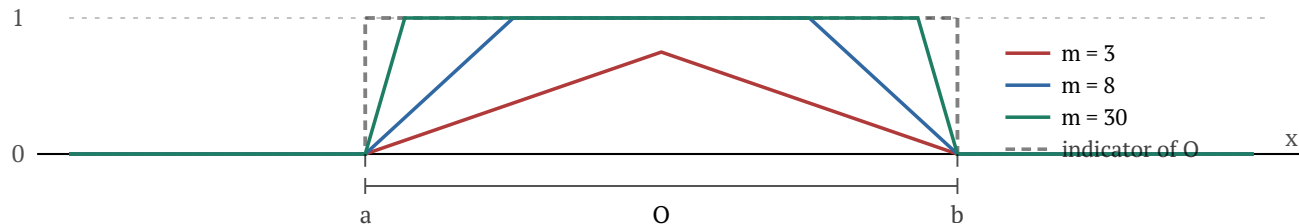
Proof, (ii) to (iii) to (v) to (vi)

(ii) implies (iii). A bounded Lipschitz function is bounded continuous.

(iii) implies (v). Let O be open and $f_m(x) = \min(1, m \text{dist}(x, O^c))$, which is m -Lipschitz, bounded by 1, and satisfies $0 \leq f_m \leq \mathbf{1}_O$ with $f_m \uparrow \mathbf{1}_O$ pointwise, since $\text{dist}(x, O^c) > 0$ exactly on O . Then

$$\liminf_n \mathbb{P}(X_n \in O) \geq \lim_n \mathbb{E}_{X_n}[f_m(X_n)] = \mathbb{E}_X[f_m(X)] \xrightarrow{m \rightarrow \infty} \mathbb{P}(X \in O)$$

by monotone convergence.



(v) implies (vi). Apply (v) to $O = F^c$ and take complements: $\limsup_n \mathbb{P}(X_n \in F) = 1 - \liminf_n \mathbb{P}(X_n \in O) \leq 1 - \mathbb{P}(X \in O) = \mathbb{P}(X \in F)$.

Proof, (v) and (vi) to (vii) and (vii) to (i)

(v) and (vi) imply (vii). Let B be Borel with $\mathbb{P}(X \in \partial B) = 0$. Then

$$\begin{aligned}
 \mathbb{P}(X \in \mathring{B}) &\leq \liminf_n \mathbb{P}(X_n \in \mathring{B}) \leq \liminf_n \mathbb{P}(X_n \in B) && \text{(v) on } \mathring{B}, \text{ then } \mathring{B} \subseteq B \\
 &\leq \limsup_n \mathbb{P}(X_n \in B) \leq \limsup_n \mathbb{P}(X_n \in \bar{B}) && B \subseteq \bar{B} \\
 &\leq \mathbb{P}(X \in \bar{B}) = \mathbb{P}(X \in \mathring{B}) && \text{(vi) on } \bar{B}, \text{ then } \mathbb{P}(X \in \partial B) = 0.
 \end{aligned}$$

The chain closes on the value it started from, so all its terms are equal.

(vii) implies (i). Take $B = (-\infty, x]$, closed, with $\mathring{B} = (-\infty, x)$ and $\partial B = \{u \leq x : u_j = x_j \text{ for some } j\}$. The events $\{X \leq x - 2^{-m} \mathbf{1}\}$ increase to $\{X < x\}$, so

$$\mathbb{P}(X \in \partial B) = F_X(x) - \mathbb{P}(X < x) = F_X(x) - \lim_{m \rightarrow \infty} F_X(x - 2^{-m} \mathbf{1}) = 0.$$

So

$$F_{X_n}(x) = \mathbb{P}(X_n \in B) \rightarrow \mathbb{P}(X \in B) = F_X(x).$$

Proof, (ii) is equivalent to (iv)

(ii) implies (iv). Let $f \geq 0$ be continuous and $M > 0$. Then $f \wedge M$ is bounded continuous and $f \geq f \wedge M$, so

$$\liminf_n \mathbb{E}_{X_n}[f(X_n)] \geq \lim_n \mathbb{E}_{X_n}[(f \wedge M)(X_n)] = \mathbb{E}_X[(f \wedge M)(X)] \xrightarrow{M \rightarrow \infty} \mathbb{E}_X[f(X)]$$

by monotone convergence.

(iv) implies (ii). Let f be bounded continuous. Both $\|f\|_\infty + f$ and $\|f\|_\infty - f$ are continuous and nonnegative, so (iv) applied to each gives

$$\liminf_n \mathbb{E}_{X_n}[f(X_n)] \geq \mathbb{E}_X[f(X)] \quad \text{and} \quad \limsup_n \mathbb{E}_{X_n}[f(X_n)] \leq \mathbb{E}_X[f(X)].$$

Exercise, differences do not converge

Claim. $X_n \xrightarrow{d} X$ does not imply $X_n - X \xrightarrow{d} 0$.

Solution. Let $U \sim \text{Unif}([0, 1])$, $X_n = U$ for every n , and $X = 1 - U$. Then $X \sim \text{Unif}([0, 1])$, so $X_n \stackrel{d}{=} X$ and in particular $X_n \xrightarrow{d} X$. But

$$X_n - X = 2U - 1 \sim \text{Unif}([-1, 1]),$$

which does not converge in distribution to 0.

What fails. Convergence in distribution controls the law of X_n and the law of X separately, never the joint law of (X_n, X) . The same remark says that $X \stackrel{d}{=} Y$ gives no information at all about (X, Z) and (Y, Z) , and it is the reason Slutsky needs one of the two sequences to have a constant limit.

Exercise, Dirac masses

Claim. $\delta_{x_n} \xrightarrow{d} \delta_x$ if and only if $x_n \rightarrow x$, for $x_n, x \in \mathbb{R}^d$.

The easy direction. If $x_n \rightarrow x$, then for every bounded continuous f ,

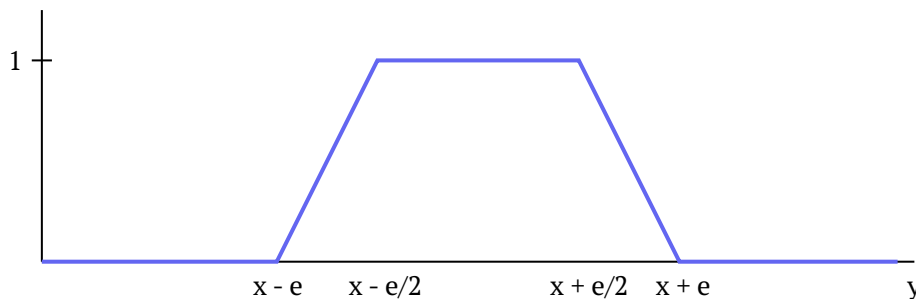
$$\mathbb{E}_{X \sim \delta_{x_n}}[f(X)] = f(x_n) \rightarrow f(x) = \mathbb{E}_{X \sim \delta_x}[f(X)]$$

by continuity of f , so portmanteau (ii) applies.

The other direction needs a test function that isolates x . It must be bounded and continuous to be admissible, so an indicator will not do.

Exercise, the isolating test function

For $\epsilon > 0$ let $f_\epsilon(y) = \min(1, (2 - \frac{2}{\epsilon}\|y - x\|)_+)$, which is continuous, bounded by 1, equal to 1 on $B(x, \epsilon/2)$ and to 0 outside $B(x, \epsilon)$.



If $\delta_{x_n} \xrightarrow{d} \delta_x$ then $f_\epsilon(x_n) \rightarrow f_\epsilon(x) = 1$ by portmanteau (ii), so $f_\epsilon(x_n) > 0$ for n large, hence $\|x_n - x\| < \epsilon$ for n large. This holds for every $\epsilon > 0$, so $x_n \rightarrow x$.

Exercise, the inequalities are strict

Claim. One sequence shows that (v) and (vi) are genuinely inequalities, and that the boundary hypothesis in (vii) cannot be dropped.

The sequence. Let $X_n \sim \mathcal{N}(0, 1/n)$ and $X \equiv 0$. For $x < 0$, $\mathbb{P}(X_n \leq x) = \mathbb{P}(\mathcal{N}(0, 1) \leq x\sqrt{n}) \rightarrow 0$, and for $x > 0$ it tends to 1, so $X_n \xrightarrow{d} X$: every $x \neq 0$ is a continuity point of F_X , and 0 is not.

(v) is strict on $O = (0, \infty)$. By symmetry $\mathbb{P}(X_n \in O) = \frac{1}{2}$ for every n , while $\mathbb{P}(X \in O) = 0$. The inequality $\liminf_n \frac{1}{2} \geq 0$ holds, with a gap of $\frac{1}{2}$.

(vi) is strict on $F = \{0\}$. Each X_n has a density so $\mathbb{P}(X_n \in F) = 0$, while $\mathbb{P}(X \in F) = 1$. Again the inequality holds and is strict, now in the other direction.

And (vii) fails on $B = (0, \infty)$. Its boundary is $\{0\}$, which carries all of the mass of X , so the hypothesis of (vii) is violated, and the conclusion fails with it: $\mathbb{P}(X_n \in B) = \frac{1}{2}$ does not tend to $\mathbb{P}(X \in B) = 0$.

Exercise, the maximum of uniforms

Claim. Let $U_1, U_2, \dots$ be independent and uniform on $[0, 1]$ and $M_n = \max_{i \leq n} U_i$. Then $n(1 - M_n) \xrightarrow{d} \mathcal{E}(1)$.

Solution. The maximum has an explicit distribution function: the U_i are independent, so for $0 \leq t \leq n$,

$$\mathbb{P}(n(1 - M_n) > t) = \mathbb{P}\left(M_n < 1 - \frac{t}{n}\right) = \prod_{i \leq n} \mathbb{P}\left(U_i < 1 - \frac{t}{n}\right) = \left(1 - \frac{t}{n}\right)^n \xrightarrow{n \rightarrow \infty} e^{-t},$$

so $\mathbb{P}(n(1 - M_n) \leq t) \rightarrow 1 - e^{-t}$ for every $t > 0$. The limit is the exponential distribution function, which is continuous everywhere, so no continuity point is excluded and the convergence in law holds.

Two remarks. The scaling is n and not $\sqrt{n}$: a maximum is not an average and the central limit theorem has nothing to say about it. And $M_n \rightarrow 1$ in probability, since $\mathbb{P}(M_n \leq 1 - \epsilon) = (1 - \epsilon)^n \rightarrow 0$, so this is the rate at which that happens.

Exercise, a dense set is enough

Claim. Let F be a distribution function and $D \subseteq \mathbb{R}$ dense. If $F_{X_n}(x) \rightarrow F(x)$ for every $x \in D$, then $X_n \xrightarrow{d} X$ with $F_X = F$.

Solution. Let x be a continuity point of F and take $u < x < v$ with $u, v \in D$, possible by density. Monotonicity of each F_{X_n} gives $F_{X_n}(u) \leq F_{X_n}(x) \leq F_{X_n}(v)$, so letting $n \rightarrow \infty$,

$$F(u) \leq \liminf_n F_{X_n}(x) \leq \limsup_n F_{X_n}(x) \leq F(v).$$

Now let $u \uparrow x$ and $v \downarrow x$ inside D . Both ends tend to $F(x)$, because F is continuous at x , and the squeeze closes.

Why it is worth stating. It is how convergence in law is checked in practice: verify the limit on the rationals, or on any convenient countable set, and the continuity points come for free. The argument is the squeeze of the portmanteau proof once more, with a dense set in place of a sequence.

Part III. Characteristic functions

The Fourier transform of a law

Definition. Let X be a random vector of $\mathbb{R}^d$. Its characteristic function, the Fourier transform of its law, is

$$\phi_X : \mathbb{R}^d \rightarrow \mathbb{C}, \quad \phi_X(t) = \mathbb{E}_X[e^{i\langle t, X \rangle}] = \int_{\mathbb{R}^d} e^{i\langle t, x \rangle} d\mathbb{P}_X(x).$$

It is defined for every t and every X , with no integrability assumption, because $|e^{i\langle t, x \rangle}| = 1$.

Why it will work. Convergence in distribution is a statement about a family of test functions indexed by all of $C_b(\mathbb{R}^d)$. The next two parts show that the family $x \mapsto e^{i\langle t, x \rangle}$ already suffices. Furthermore, the Fourier transform enjoys many nice properties.

Elementary properties

Proposition. Let X, Y be random vectors of $\mathbb{R}^d$.

(i) $\phi_X = \phi_Y \iff \mathbb{P}_X = \mathbb{P}_Y \iff X \stackrel{d}{=} Y$.

(ii) $|\phi_X(t)| \leq 1$ for every t , and $\phi_X(0) = 1$.

(iii) ϕ_X is uniformly continuous on $\mathbb{R}^d$.

(iv) For $A \in \mathbb{R}^{d \times d}$ and $b \in \mathbb{R}^d$, $\phi_{AX+b}(t) = \phi_X(A^\top t) e^{i\langle b, t \rangle}$.

(v) $\phi_{-X}(t) = \phi_X(-t) = \overline{\phi_X(t)}$, so X is symmetric if and only if ϕ_X is real valued.

Proof of (i), the smoothing identity

(i) The implication $\Leftarrow$ is immediate. For the converse, we need the following lemma

Lemma. Let $Z \sim \mathcal{N}(0, I_d)$ be independent of X , let $\sigma > 0$, and let q_σ be the density of $\mathcal{N}(0, \sigma^2 I_d)$. Then $X + \sigma Z$ has the density

$$p_\sigma^X(x) = \mathbb{E}_X[q_\sigma(x - X)] = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi_X(t) e^{-\sigma^2 \|t\|^2/2} dt.$$

$$\begin{aligned} \mathbb{E}_X[q_\sigma(x - X)] &= \int_{\mathbb{R}^d} q_\sigma(x - y) d\mathbb{P}_X(y) && \text{definition of a convolution} \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} (2\pi)^{-d} e^{i\langle t, y-x \rangle} e^{-\sigma^2 \|t\|^2/2} dt d\mathbb{P}_X(y) && \text{gaussian transform, next part} \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} (2\pi)^{-d} e^{-i\langle t, x \rangle} e^{i\langle t, y \rangle} e^{-\sigma^2 \|t\|^2/2} d\mathbb{P}_X(y) dt && \text{Fubini, modulus } e^{-\sigma^2 \|t\|^2/2} \\ &= (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi_X(t) e^{-\sigma^2 \|t\|^2/2} dt && \text{definition of } \phi_X. \quad \blacksquare \end{aligned}$$

Proof of (i), removing the smoothing

Suppose $\phi_X = \phi_Y$. By the lemma, $X + \sigma Z$ and $Y + \sigma Z$ have the same density for every $\sigma > 0$, hence the same law.

Let f be bounded continuous and take $\sigma_n = 1/n$, dominated convergence being a statement about sequences.

For almost every ω the vector $Z(\omega)$ is finite, so $X(\omega) + \sigma_n Z(\omega) \rightarrow X(\omega)$, and f is continuous, so $f(X + \sigma_n Z) \rightarrow f(X)$ almost surely.

$|f(X + \sigma_n Z)| \leq \|f\|_\infty$ for every n , and the constant $\|f\|_\infty$ is integrable because $\mathbb{P}$ is a probability measure. Both facts hold with Y in place of X , so by dominated convergence

$$\mathbb{E}_X[f(X)] = \lim_{n \rightarrow \infty} \mathbb{E}_{X,Z}[f(X + \sigma_n Z)] = \lim_{n \rightarrow \infty} \mathbb{E}_{Y,Z}[f(Y + \sigma_n Z)] = \mathbb{E}_Y[f(Y)],$$

the middle equality holding term by term by the first step.

Proof of (i), separating the laws

Bounded continuous functions now determine the law.

For F closed and nonempty, note $f_m(x) = (1 - m \operatorname{dist}(x, F))_+$ for $m \in \mathbb{N}$. Each f_m is bounded by 1, is m -Lipschitz, equals 1 on F , and decreases pointwise to $\mathbf{1}_F$: if $x \notin F$ then $\operatorname{dist}(x, F) > 0$ because F is closed, so $f_m(x) = 0$ as soon as $m > 1/\operatorname{dist}(x, F)$.

Dominated convergence once more, with the constant 1 as dominating function, gives

$$\mathbb{P}(X \in F) = \lim_{m \rightarrow \infty} \mathbb{E}_X[f_m(X)] = \lim_{m \rightarrow \infty} \mathbb{E}_Y[f_m(Y)] = \mathbb{P}(Y \in F).$$

Closed sets form a π -system generating the Borel σ -algebra, so two probability measures agreeing on all of them are equal, and $\mathbb{P}_X = \mathbb{P}_Y$. ■

Proof of the elementary properties

(ii) $\phi_X(0) = \mathbb{E}_X[1] = 1$ and $|\phi_X(t)| = |\mathbb{E}_X[e^{i\langle t, X \rangle}]| \leq \mathbb{E}_X[|e^{i\langle t, X \rangle}|] = 1$.

(iii) Let $\epsilon > 0$ and pick M with $\mathbb{P}(X \notin B(0, M)) \leq \epsilon$. Split the expectation at $B(0, M)$: outside, $|e^{ia} - e^{ib}| \leq 2$ and the mass is at most ϵ ; inside, $|e^{ia} - e^{ib}| \leq |a - b|$ (mean value inequality) and $|\langle t_1 - t_2, x \rangle| \leq M\|t_1 - t_2\|$ by Cauchy and Schwarz. Hence

$$|\phi_X(t_1) - \phi_X(t_2)| \leq \int_{B(0, M)} |e^{i\langle t_1, x \rangle} - e^{i\langle t_2, x \rangle}| d\mathbb{P}_X(x) + 2\epsilon \leq M\|t_1 - t_2\| + 2\epsilon.$$

So $\|t_1 - t_2\| \leq \epsilon/M$ gives $|\phi_X(t_1) - \phi_X(t_2)| \leq 3\epsilon$ for every such pair.

(iv) and (v) are direct: $\langle t, Ax + b \rangle = \langle A^\top t, x \rangle + \langle b, t \rangle$, and $\overline{e^{iu}} = e^{-iu}$.

Moments, densities, independence

Proposition, continued. With $X^\alpha = \prod_{j \leq d} X_j^{\alpha_j}$ for a multi-index α and $|\alpha| = \sum_j \alpha_j$,

(vi) if $\mathbb{E}_X[\|X\|^r] < \infty$ then $\phi_X \in C^r$, and $\partial^\alpha \phi_X(t) = i^{|\alpha|} \mathbb{E}_X[X^\alpha e^{i\langle t, X \rangle}]$ for $|\alpha| \leq r$.

(vii) the converse is false, but if ϕ_X has a derivative of even order $2r$ at 0 then $\mathbb{E}_X[\|X\|^{2r}] < \infty$.

(viii) if X has a density with respect to Lebesgue measure then $\phi_X(t) \rightarrow 0$ as $\|t\| \rightarrow \infty$.

(ix) if $\phi_X \in L^1$ then X has a bounded continuous density, given by the inversion formula $p(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi_X(t) dt$.

(x) X and Y are independent if and only if $\phi_{(X,Y)}(s, t) = \phi_X(s)\phi_Y(t)$ for all (s, t) .

(xi) if X and Y are independent then $\mathbb{P}_{X+Y} = \mathbb{P}_X * \mathbb{P}_Y$ and $\phi_{X+Y} = \phi_X \phi_Y$.

Proof, differentiability and moments

(vi) Differentiation under the integral sign. The candidate derivative $x \mapsto ix_j e^{i\langle t, x \rangle}$ is dominated by $|x_j| \leq \|x\|$, which is integrable when $r \geq 1$, uniformly in t . Iterate r times, the α -th derivative being dominated by $\|x\|^{|\alpha|}$.

(vii) Take $d = 1$ and $r = 1$. If $\phi_X''(0)$ exists then it equals the limit of the symmetric second difference, and $\phi_X(t) + \phi_X(-t) = 2\mathbb{E}_X[\cos(tX)]$, so

$$\phi_X''(0) = \lim_{t \rightarrow 0} \frac{\phi_X(t) + \phi_X(-t) - 2\phi_X(0)}{t^2} = -2 \lim_{t \rightarrow 0} \mathbb{E}_X \left[\frac{1 - \cos(tX)}{t^2} \right].$$

Since $(1 - \cos(tx))/t^2 \rightarrow x^2/2$ as $t \rightarrow 0$ and the integrand is nonnegative, Fatou along any sequence $t_n \rightarrow 0$ gives

$$\mathbb{E}_X[X^2] = \mathbb{E}_X \left[\liminf_n 2 \frac{1 - \cos(t_n X)}{t_n^2} \right] \leq 2 \liminf_n \mathbb{E}_X \left[\frac{1 - \cos(t_n X)}{t_n^2} \right] = -\phi_X''(0) < \infty.$$

For higher orders the reasoning is the same.

(viii), The Riemann and Lebesgue lemma

(viii) Item (viii) says nothing about probability. It is a statement about integrable functions, which the density then happens to satisfy.

Lemma (Riemann and Lebesgue). For $f \in L^1(\mathbb{R}^d)$ set $\mathcal{F}f(t) = \int_{\mathbb{R}^d} f(x) e^{i\langle t, x \rangle} dx$. Then $\mathcal{F}f(t) \rightarrow 0$ as $\|t\| \rightarrow \infty$.

Proof. Let $t \neq 0$ and put $h = \pi t / \|t\|^2$, so that $\langle t, h \rangle = \pi$ and $\|h\| = \pi / \|t\|$. Substituting $x = y + h$ and using $e^{i\pi} = -1$,

$$\mathcal{F}f(t) = \int f(x) e^{i\langle t, x \rangle} dx = e^{i\langle t, h \rangle} \int f(y + h) e^{i\langle t, y \rangle} dy = - \int f(y + h) e^{i\langle t, y \rangle} dy.$$

Averaging this with the original expression, the oscillating factor cancels:

$$2\mathcal{F}f(t) = \int (f(y) - f(y + h)) e^{i\langle t, y \rangle} dy, \quad |\mathcal{F}f(t)| \leq \frac{1}{2} \|f - f(\cdot + h)\|_{L^1}.$$

Proof of the lemma, translation in L^1

Since $\|h\| = \pi/\|t\|$, it remains to prove that $\|f - f(\cdot + h)\|_{L^1} \rightarrow 0$ as $h \rightarrow 0$.

For g continuous with compact support, all the functions $g(\cdot + h)$ with $\|h\| \leq 1$ vanish off one fixed compact set K , and g is uniformly continuous, so $\|g - g(\cdot + h)\|_{L^1} \leq \lambda(K) \sup_{x \in \mathbb{R}^d} |g(x) - g(x + h)| \rightarrow 0$, with λ the Lebesgue measure.

For general f , such g are dense in $L^1(\mathbb{R}^d)$, see Rudin, *Real and Complex Analysis*, Theorem 3.14. Given $\epsilon > 0$ pick g with $\|f - g\|_{L^1} < \epsilon/3$. Translation preserves the L^1 norm, so

$$\|f - f(\cdot + h)\|_{L^1} \leq 2\|f - g\|_{L^1} + \|g - g(\cdot + h)\|_{L^1} < \epsilon$$

for $\|h\|$ small enough. ■

Item (viii) is now one line. If X has a density p with respect to Lebesgue measure then $p \in L^1(\mathbb{R}^d)$ and $\phi_X = \mathcal{F}p$, so $\phi_X(t) \rightarrow 0$ as $\|t\| \rightarrow \infty$. ■

(ix), The Fourier inversion theorem

(ix) Theorem (Fourier inversion). Let X be a random vector of $\mathbb{R}^d$ with $\phi_X \in L^1(\mathbb{R}^d)$. Then $\mathbb{P}_X$ has a bounded continuous density with respect to Lebesgue measure, namely

$$p(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi_X(t) dt.$$

We write $\phi = \phi_X$. Fubini does not apply to the inversion integral directly, so we smooth it first: for $\sigma > 0$ set

$$p_\sigma(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi(t) e^{-\sigma^2 \|t\|^2/2} dt = \mathbb{E}_X[q_\sigma(x - X)],$$

the second equality being the lemma proved for (i), where $Z \sim \mathcal{N}(0, I_d)$ is independent of X and q_σ is the density of $\mathcal{N}(0, \sigma^2 I_d)$. So p_σ is the density of $X + \sigma Z$.

Proof of the theorem, the transform side

The first step is that $p_\sigma \rightarrow p$ uniformly on $\mathbb{R}^d$.

$$\begin{aligned} |p_\sigma(x) - p(x)| &= (2\pi)^{-d} \left| \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi(t) \left(e^{-\sigma^2 \|t\|^2/2} - 1 \right) dt \right| && \text{definitions of } p_\sigma \text{ and } p \\ &\leq (2\pi)^{-d} \int_{\mathbb{R}^d} |\phi(t)| \left(1 - e^{-\sigma^2 \|t\|^2/2} \right) dt && \text{triangle inequality} \\ &\xrightarrow[\sigma \rightarrow 0]{} 0 && \text{dominated by } |\phi| \in L^1 \end{aligned}$$

The last bound is free of x , which is the uniformity. The same domination gives $\|p\|_\infty \leq (2\pi)^{-d} \|\phi\|_{L^1}$ and, by dominated convergence in x , continuity of p .

Proof of the theorem, the law side

Let $R = (a, b]$ be a bounded rectangle with $\mathbb{P}(X \in \partial R) = 0$.

$$\begin{aligned}\mathbb{P}(X \in R) &= \lim_{\sigma \rightarrow 0} \mathbb{P}(X + \sigma Z \in R) \quad \sigma Z \rightarrow 0 \text{ a.s. and } \mathbb{P}(X \in \partial R) = 0 \\ &= \lim_{\sigma \rightarrow 0} \int_R p_\sigma(x) dx \quad p_\sigma \text{ is the density of } X + \sigma Z \\ &= \int_R p(x) dx \quad p_\sigma \rightarrow p \text{ uniformly, } R \text{ bounded}\end{aligned}$$

Such rectangles form a π -system generating the Borel σ -algebra, since for each coordinate all but countably many levels carry no mass. Two probability measures agreeing on it are equal, so p is a density of $\mathbb{P}_X$. ■

Corollary, inversion in L^1 . If $f \in L^1(\mathbb{R}^d)$ is a probability density with $\mathcal{F}f \in L^1(\mathbb{R}^d)$, then $f = p$ almost everywhere: a density whose transform is integrable has a bounded continuous version.

Proof, independence and sums

(x) If X and Y are independent, Fubini gives $\mathbb{E}_{X,Y}[e^{i\langle s,X\rangle+i\langle t,Y\rangle}] = \phi_X(s)\phi_Y(t)$. Conversely, the right side is the characteristic function of $\mathbb{P}_X \otimes \mathbb{P}_Y$, so by (i) the joint law equals the product law, which is independence.

(xi) Take $s = t$ in (x), or directly $\mathbb{E}_{X,Y}[e^{i\langle t,X+Y\rangle}] = \mathbb{E}_X[e^{i\langle t,X\rangle}]\mathbb{E}_Y[e^{i\langle t,Y\rangle}]$ by Fubini.

The standard gaussian, by a differential equation

Proposition. Let $X \sim \mathcal{N}(0, 1)$ on $\mathbb{R}$. Then $\phi_X(t) = e^{-t^2/2}$ for every $t \in \mathbb{R}$, so the standard gaussian is its own Fourier transform up to the normalising constant.

Three proofs follow: a differential equation here, analytic continuation and a contour integral next. The object throughout is $\phi_X(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{itx} e^{-x^2/2} dx$, and the plan here is to show that it solves $\phi'_X = -t\phi_X$, whose solutions are the multiples of $e^{-t^2/2}$. By (vi) with $r = 1$,

$$\begin{aligned}\phi'_X(t) &= \frac{i}{\sqrt{2\pi}} \int_{\mathbb{R}} \underbrace{e^{itx}}_{\text{easy to differentiate}} \underbrace{xe^{-x^2/2}}_{\text{easy to integrate}} dx && \text{differentiation under the integral} \\ &= \frac{-i}{\sqrt{2\pi}} \int_{\mathbb{R}} ite^{itx} e^{-x^2/2} dx = -t\phi_X(t) && \text{integration by parts, boundary terms vanish}\end{aligned}$$

So $\phi'_X + t\phi_X = 0$, hence $\phi_X(t) = \phi_X(0)e^{-t^2/2}$, and $\phi_X(0) = 1$ by (ii). Therefore

$$\phi_X(t) = e^{-t^2/2}.$$

The standard gaussian, by analytic continuation

Set $F(z) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{zx} e^{-x^2/2} dx$ for $z \in \mathbb{C}$, so that $\phi_X(t) = F(it)$. The integrand is holomorphic in z and locally dominated by $e^{R|x|} e^{-x^2/2}$ on $\{|z| \leq R\}$, so F is entire.

For $z = s$ real, complete the square, $sx - \frac{x^2}{2} = -\frac{(x-s)^2}{2} + \frac{s^2}{2}$, and use the gaussian integral:

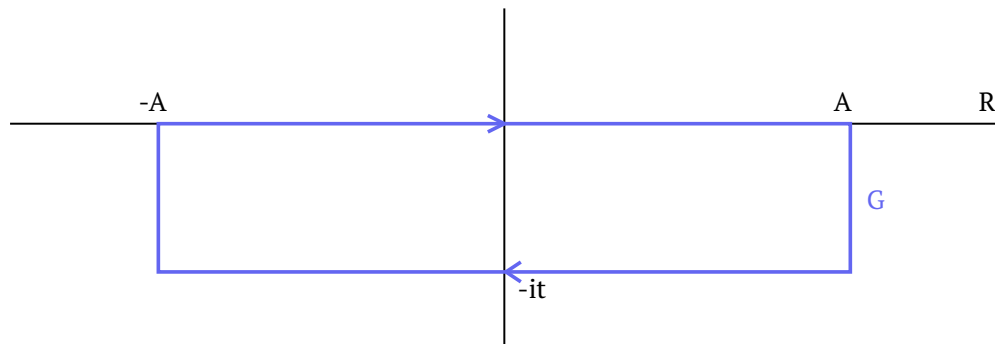
$$F(s) = \frac{e^{s^2/2}}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(x-s)^2/2} dx = e^{s^2/2}.$$

The entire functions F and $z \mapsto e^{z^2/2}$ agree on $\mathbb{R}$, which has an accumulation point, so they agree on $\mathbb{C}$. Hence $\phi_X(t) = F(it) = e^{-t^2/2}$.

The standard gaussian, by a contour integral

Complete the square inside the exponential: $itx - \frac{x^2}{2} = -\frac{(x-it)^2}{2} - \frac{t^2}{2}$, so it suffices to prove

$\int_{\mathbb{R}} e^{-(x-it)^2/2} dx = \sqrt{2\pi}$. Take $t > 0$, the case $t < 0$ following from property (v). Apply Cauchy's theorem to the entire function $g(u) = e^{-u^2/2}$ on the rectangle Γ_A with vertices $\pm A$ and $\pm A - it$.



The two vertical sides have length $|t|$ and integrand bounded by $e^{-(A^2-t^2)/2}$, so they vanish as $A \rightarrow \infty$. The top side tends to $\int_{\mathbb{R}} e^{-x^2/2} dx = \sqrt{2\pi}$ and the bottom to $\int_{\mathbb{R}} e^{-(x-it)^2/2} dx$, both by dominated convergence. Since the total is 0, the two limits are equal.

The gaussian in $\mathbb{R}^d$

Isotropic case. If $X \sim \mathcal{N}(0, I_d)$ the coordinates are independent standard gaussians, so by (x) and the scalar computation,

$$\phi_X(t) = \prod_{j \leq d} \phi_{X_j}(t_j) = \prod_{j \leq d} e^{-t_j^2/2} = e^{-\|t\|^2/2}.$$

General case. Let $\Sigma \succeq 0$, diagonalise $\Sigma = P^\top D P$ with P orthogonal and $D = \text{diag}(\lambda_1, \dots, \lambda_d)$, and set $\Sigma^{1/2} = P^\top D^{1/2} P$. If $X' \sim \mathcal{N}(0, I_d)$ then $X = \Sigma^{1/2} X' + \mu \sim \mathcal{N}(\mu, \Sigma)$, and (iv) gives

$$\phi_X(t) = e^{i\langle \mu, t \rangle} e^{-\|\Sigma^{1/2} t\|^2/2} = \exp(i\langle \mu, t \rangle - \tfrac{1}{2} t^\top \Sigma t).$$

Theorem. Let $\mu \in \mathbb{R}^d$ and let Σ be symmetric positive semidefinite, possibly singular. Then $X \sim \mathcal{N}(\mu, \Sigma)$ if and only if

$$\phi_X(t) = \exp \left(i\langle \mu, t \rangle - \tfrac{1}{2} t^\top \Sigma t \right), \quad t \in \mathbb{R}^d.$$

Part IV. Tightness, Prohorov, Levy

Uniform tightness

Definition. A family $(X_a)_{a \in A}$ of random vectors of $\mathbb{R}^d$ is uniformly tight if

$$\forall \epsilon > 0, \exists M > 0 \text{ such that } \sup_{a \in A} \mathbb{P}(\|X_a\| \geq M) \leq \epsilon,$$

equivalently

$$\sup_{a \in A} \mathbb{P}(\|X_a\| \geq M) \xrightarrow{M \rightarrow \infty} 0.$$

A single random vector is tight, since $\{\|X\| \geq M\} \downarrow \emptyset$ as $M \uparrow \infty$, so $\mathbb{P}(\|X\| \geq M) \rightarrow 0$ by monotone convergence. A finite family is tight, by taking the largest of the finitely many M .

Reading. Tightness says no mass escapes to infinity. It is a property of the laws $\mathbb{P}_{X_a}$, not of the random vectors, and it is the analogue of boundedness for a sequence of real numbers.

Prohorov's theorem

Theorem. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random vectors of $\mathbb{R}^d$.

1. If $X_n \xrightarrow{d} X$ for some random vector X , then $(X_n)_n$ is uniformly tight.
2. If $(X_n)_n$ is uniformly tight then there exist a random vector X and a subsequence $(X_{\varphi(n)})$ with $X_{\varphi(n)} \xrightarrow{d} X$.

The analogy. In $\mathbb{R}$: a convergent sequence is bounded, and from a bounded sequence one can extract a convergent subsequence. Here: a sequence convergent in distribution is tight, and from a tight sequence one can extract a subsequence convergent in distribution. This allows compactness-like arguments.

Proof of Prohorov, first item

Let $\epsilon > 0$. Since X is tight there is M with $\mathbb{P}(\|X\| \geq M) < \epsilon$. The set $\{x : \|x\| \geq M\}$ is closed, so portmanteau (vi) gives

$$\limsup_n \mathbb{P}(\|X_n\| \geq M) \leq \mathbb{P}(\|X\| \geq M) < \epsilon.$$

Hence there is N with $\mathbb{P}(\|X_n\| \geq M) < 2\epsilon$ for every $n \geq N$. The finitely many indices $n < N$ are handled by enlarging M , each of them being tight on its own. So $\sup_n \mathbb{P}(\|X_n\| \geq M') < 2\epsilon$ for some $M' \geq M$, and ϵ was arbitrary.

Remark. The two steps are exactly the two halves of "a convergent sequence is bounded": the tail is controlled by the limit, the head is finite.

Helly's selection lemma

Lemma (Helly). From any sequence (F_n) of cumulative distribution functions on $\mathbb{R}^d$ one can extract a subsequence converging, at every continuity point, to a nondecreasing right continuous $F : \mathbb{R}^d \rightarrow [0, 1]$.

Proof. Diagonal extraction. Enumerate $\mathbb{Q}^d$ as $q^{(1)}, q^{(2)}, \dots$, which is possible because it is countable. The sequence $(F_n(q^{(1)}))_n$ lies in $[0, 1]$, hence has a convergent subsequence; extract again along it for $q^{(2)}$, and so on. The diagonal subsequence φ , whose n -th term is the n -th term of the n -th extraction, satisfies $F_{\varphi(n)}(q) \rightarrow G(q)$ for every $q \in \mathbb{Q}^d$, and G is nondecreasing as a pointwise limit of nondecreasing functions.

The candidate limit. Write $q > x$ for $q_j > x_j$ at every coordinate j , and set

$$F(x) = \inf \{ G(q) : q \in \mathbb{Q}^d, q > x \} \in [0, 1].$$

If $x \leq y$ then $\{q > y\} \subset \{q > x\}$, an infimum over a smaller set, so $F(x) \leq F(y)$.

Proof of Helly, right continuity and convergence

Right continuity. Let $x^{(m)} \downarrow x$ coordinatewise. Monotonicity gives $F(x^{(m)}) \downarrow L \geq F(x)$. Given $\epsilon > 0$ pick $q \in \mathbb{Q}^d$ with $q > x$ and $G(q) \leq F(x) + \epsilon$, possible by definition of the infimum. For m large $x^{(m)} < q$, so q competes in the infimum defining $F(x^{(m)})$ and $F(x^{(m)}) \leq G(q) \leq F(x) + \epsilon$. Hence $L = F(x)$.

Convergence at a continuity point x . Let $\epsilon > 0$. Continuity of F at x gives $y < x$ with $F(y) \geq F(x) - \epsilon$. Pick $r \in \mathbb{Q}^d$ with $y < r < x$ and $q \in \mathbb{Q}^d$ with $q > x$ and $G(q) \leq F(x) + \epsilon$. Then

$$\begin{aligned} \liminf_n F_{\varphi(n)}(x) &\geq \lim_n F_{\varphi(n)}(r) = G(r) \geq F(y) \geq F(x) - \epsilon & r < x, r > y \\ \limsup_n F_{\varphi(n)}(x) &\leq \lim_n F_{\varphi(n)}(q) = G(q) \leq F(x) + \epsilon & x < q \end{aligned}$$

so $F_{\varphi(n)}(x) \rightarrow F(x)$. ■

Proof of Prohorov, second item

Apply Helly to $F_n = F_{X_n}$ and let F be the limit along φ .

The limit may be defective, meaning F need not tend to 1 as all coordinates tend to $+\infty$: mass can escape. Tightness forbids this. Given $\epsilon > 0$, pick M with $\sup_n \mathbb{P}(\|X_n\| \geq M) \leq \epsilon$. Then, along the extracted subsequence and at continuity points,

$$x_j > M \ \forall j \implies F(x) = \lim_n F_n(x) \geq 1 - \epsilon, \quad x_j < -M \text{ for some } j \implies F(x) \leq \epsilon.$$

So F tends to 1 and to 0 at the two ends, hence is a genuine cumulative distribution function, and it is the one of some random vector X with $X_{\varphi(n)} \xrightarrow{d} X$.

Levy's continuity theorem

Theorem. Let (X_n) and X be random vectors of $\mathbb{R}^d$.

1. If $X_n \xrightarrow{d} X$ then $\phi_{X_n}(t) \rightarrow \phi_X(t)$ for every $t \in \mathbb{R}^d$.
2. If there is $\phi : \mathbb{R}^d \rightarrow \mathbb{C}$, continuous at 0, with $\phi_{X_n}(t) \rightarrow \phi(t)$ for every t , then ϕ is the characteristic function of some random vector X and $X_n \xrightarrow{d} X$.

The asymmetry is essential. In item 2 the limit ϕ is not assumed to be a characteristic function, only continuous at the single point 0. Without that assumption the conclusion is false: for $X_n \sim \mathcal{N}(0, n)$, $\phi_{X_n}(t) = e^{-nt^2/2} \rightarrow \mathbf{1}\{t = 0\}$ pointwise, a limit discontinuous at 0, and indeed X_n does not converge in distribution.

Proof of Levy, first item

The functions $x \mapsto \cos\langle t, x \rangle$ and $x \mapsto \sin\langle t, x \rangle$ are bounded and continuous, so portmanteau (ii) applies to each and

$$\phi_{X_n}(t) = \mathbb{E}_{X_n}[\cos\langle t, X_n \rangle] + i\mathbb{E}_{X_n}[\sin\langle t, X_n \rangle] \rightarrow \mathbb{E}_X[\cos\langle t, X \rangle] + i\mathbb{E}_X[\sin\langle t, X \rangle] = \phi_X(t).$$

Proof of Levy, second item, at most one limit point

Suppose (X_n) is uniformly tight and that two subsequences converge in distribution, $X_{\varphi_1(n)} \xrightarrow{d} X$ and $X_{\varphi_2(n)} \xrightarrow{d} X'$. By item 1,

$$\phi_X(t) = \lim_n \phi_{X_{\varphi_1(n)}}(t) = \phi(t) = \lim_n \phi_{X_{\varphi_2(n)}}(t) = \phi_{X'}(t) \quad \text{for every } t,$$

so $X \stackrel{d}{=} X'$ by injectivity, property (i).

A tight sequence with a single limit point converges. If $X_n \not\xrightarrow{d} X$, there are $\epsilon > 0$, a bounded continuous f and a subsequence along which $|\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| \geq \epsilon$. That subsequence is tight, so by Prohorov it has a further subsequence converging in distribution, necessarily to X by the paragraph above, contradicting the separation.

So item 2 is proved as soon as (X_n) is shown to be uniformly tight.

Proof of Levy, tightness from continuity at zero

Key inequality. For a real random variable Z and $\delta > 0$, by Fubini and $\frac{1}{\delta} \int_{-\delta}^{\delta} (1 - e^{itz}) dt = 2 - \frac{2 \sin(\delta z)}{\delta z}$,

$$\frac{1}{\delta} \int_{-\delta}^{\delta} (1 - \operatorname{Re} \phi_Z(t)) dt = 2 \mathbb{E}_Z \left[1 - \frac{\sin(\delta Z)}{\delta Z} \right] \geq \mathbb{P}(|Z| \geq 2/\delta),$$

because $1 - \frac{\sin u}{u} \geq 0$ for all u , and $1 - \frac{\sin u}{u} \geq \frac{1}{2}$ as soon as $|u| \geq 2$.

Apply this to $Z = \langle e_j, X_n \rangle$, whose characteristic function is $s \mapsto \phi_{X_n}(se_j)$. The integrand is bounded by 2, so dominated convergence gives

$$\limsup_n \mathbb{P}(|X_{n,j}| \geq 2/\delta) \leq \frac{1}{\delta} \int_{-\delta}^{\delta} (1 - \operatorname{Re} \phi(se_j)) ds,$$

and the right side tends to 0 as $\delta \rightarrow 0$ since ϕ is continuous at 0 with $\phi(0) = \lim_n \phi_{X_n}(0) = 1$.

Each coordinate is therefore tight, hence so is (X_n) , and the previous slide concludes.

The Cramer and Wold device

Corollary. $X_n \xrightarrow{d} X$ in $\mathbb{R}^d$ if and only if $\langle t, X_n \rangle \xrightarrow{d} \langle t, X \rangle$ in $\mathbb{R}$ for every $t \in \mathbb{R}^d$.

Proof. For $s \in \mathbb{R}$, $\phi_{\langle t, X_n \rangle}(s) = \mathbb{E}_{X_n}[e^{is\langle t, X_n \rangle}] = \phi_{X_n}(st)$. If all one-dimensional projections converge in distribution, then taking $s = 1$ gives $\phi_{X_n}(t) \rightarrow \phi_X(t)$ for every t , and ϕ_X is continuous, so Levy item 2 applies. The converse is Levy item 1 read at st .

Why it is used. It reduces every multivariate limit theorem to its scalar version at no cost. The multivariate central limit theorem in a few slides is proved this way.

Example, the weak law of large numbers

Proposition. Let $X_1, \dots, X_n$ be i.i.d. copies of a real X whose characteristic function is differentiable at 0 with $\phi'_X(0) = i\mu$, and let $\bar{X}_n = \frac{1}{n} \sum_{i \leq n} X_i$. Then $\bar{X}_n \xrightarrow{d} \mu$, and hence $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$ by the theorem on constant limits in Part V.

Proof. By independence, property (xi), and differentiability at 0 with $\phi_X(0) = 1$,

$$\phi_{\bar{X}_n}(t) = \phi_X\left(\frac{t}{n}\right)^n = \left(1 + \frac{it\mu}{n} + o\left(\frac{1}{n}\right)\right)^n \xrightarrow{n \rightarrow \infty} e^{it\mu} = \phi_{\delta_\mu}(t),$$

which is continuous, so Levy item 2 applies.

Remarks on the law of large numbers

If $X \in L^1$ then ϕ_X is C^1 with $\phi'_X(0) = i\mathbb{E}_X[X]$ by property (vi), so the proposition applies with $\mu = \mathbb{E}_X[X]$ and recovers the usual weak law.

Theorem (strong law of large numbers). Let $(X_i)_{i \in \mathbb{N}}$ be i.i.d. copies of a random vector X of $\mathbb{R}^d$ with $\mathbb{E}_X[\|X\|] < \infty$, and let $\bar{X}_n = \frac{1}{n} \sum_{i \leq n} X_i$. Then $\bar{X}_n \xrightarrow{a.s.} \mathbb{E}_X[X]$.

Proof omitted. This is strictly stronger than the weak law. It needs a truncation and a Borel and Cantelli argument, not a Fourier argument, since almost sure convergence is not a property of the laws alone. See van der Vaart, Proposition 2.16.

What the Fourier proof does and does not see. It manipulates $\mathbb{P}_{\bar{X}_n}$ only, so it can never produce an almost sure statement. That is the structural reason the two laws of large numbers have unrelated proofs.

Example, the central limit theorem

Theorem. Let $X_1, \dots, X_n$ be i.i.d. copies of a real X with $\mathbb{E}_X[X] = 0$ and $\mathbb{E}_X[X^2] = 1$. Then $\sqrt{n}\bar{X}_n \xrightarrow{d} \mathcal{N}(0, 1)$.

Proof. By property (vi) with $r = 2$, ϕ_X is C^2 with $\phi_X(0) = 1$, $\phi'_X(0) = i\mathbb{E}_X[X] = 0$ and $\phi''_X(0) = -\mathbb{E}_X[X^2] = -1$, so $\phi_X(u) = 1 - \frac{u^2}{2} + o(u^2)$ near 0. Hence

$$\phi_{\sqrt{n}\bar{X}_n}(t) = \phi_X\left(\frac{t}{\sqrt{n}}\right)^n = \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right)^n \xrightarrow{n \rightarrow \infty} e^{-t^2/2},$$

the characteristic function of $\mathcal{N}(0, 1)$, which is continuous. Levy item 2 concludes.

Where each hypothesis enters. Centering kills the first order term, unit variance fixes the scale.

The central limit theorem in $\mathbb{R}^d$

Theorem. Let $X_1, \dots, X_n$ be i.i.d. copies of a random vector X of $\mathbb{R}^d$ with $\mathbb{E}_X[\|X\|^2] < \infty$, mean μ and covariance $\Sigma = \text{Cov}_X(X)$. Then

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \Sigma).$$

Proof. Fix $t \in \mathbb{R}^d$ and apply the scalar theorem to the i.i.d. real variables $\langle t, X_i \rangle$, which have mean $\langle t, \mu \rangle$ and variance $t^\top \Sigma t$:

$$\langle t, \sqrt{n}(\bar{X}_n - \mu) \rangle = \sqrt{n} \left(\frac{1}{n} \sum_{i \leq n} \langle t, X_i \rangle - \langle t, \mu \rangle \right) \xrightarrow{d} \mathcal{N}(0, t^\top \Sigma t) \stackrel{d}{=} \langle t, \mathcal{N}(0, \Sigma) \rangle.$$

Every projection converges, so Cramer and Wold gives the statement.

Remark. If $t^\top \Sigma t = 0$ for some $t \neq 0$ the projected limit is δ_0 , and the limiting gaussian is degenerate. Nothing breaks: $\mathcal{N}(0, \Sigma)$ was defined through its characteristic function.

Part V. Relations between the modes of convergence

The chain of implications

Theorem. For random vectors on a common probability space,

$$X_n \xrightarrow{a.s.} X \implies X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X, \quad X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{\mathbb{P}} X.$$

Proof of the first. Fix $\epsilon > 0$. For almost every ω , $\|X_n(\omega) - X(\omega)\| \rightarrow 0$, so $\mathbf{1}\{\|X_n(\omega) - X(\omega)\| \geq \epsilon\} \rightarrow 0$. The indicators are bounded by 1, so dominated convergence gives

$$\mathbb{P}(\|X_n - X\| \geq \epsilon) = \int_{\Omega} \mathbf{1}\{\|X_n - X\| \geq \epsilon\} d\mathbb{P} \rightarrow 0.$$

Proof of the third. Markov applied to $\|X_n - X\|^p$: $\mathbb{P}(\|X_n - X\| \geq \epsilon) \leq \epsilon^{-p} \mathbb{E}_{X_n, X}[\|X_n - X\|^p] \rightarrow 0$.

Convergence in probability implies convergence in law

Let f be bounded by M and L -Lipschitz, and let $\epsilon > 0$. Split on whether X_n is close to X :

$$\begin{aligned} & |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| \\ & \leq \mathbb{E}_{X_n, X}[|f(X_n) - f(X)|] && \text{Jensen} \\ & \leq \mathbb{E}_{X_n, X}[L\epsilon \mathbf{1}\{\|X_n - X\| \leq \epsilon\} + 2M\mathbf{1}\{\|X_n - X\| \geq \epsilon\}] && \text{Lipschitz, then bounded} \\ & \leq L\epsilon + 2M\mathbb{P}(\|X_n - X\| \geq \epsilon) && \text{monotonicity} \end{aligned}$$

The second term tends to 0, so $\limsup_n |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| \leq L\epsilon$ for every $\epsilon > 0$, hence the limsup is 0. Portmanteau (iii) concludes.

None of the implications reverses

In probability but not almost surely. On $\Omega = [0, 1]$ with Lebesgue measure, take the typewriter sequence: for $n = 2^k + j$ with $0 \leq j < 2^k$, let $X_n = \mathbf{1}_{[j2^{-k}, (j+1)2^{-k}]}$. Then $\mathbb{P}(X_n \neq 0) = 2^{-k} \rightarrow 0$, yet for every ω the sequence $X_n(\omega)$ takes the value 1 infinitely often and the value 0 infinitely often.

Almost surely but not in L^1 . $X_n = n\mathbf{1}_{[0, 1/n]}$ on the same space: $X_n \rightarrow 0$ almost surely, but $\mathbb{E}_{X_n}[X_n] = 1$ for every n .

In law but not in probability. $U \sim \text{Unif}([0, 1])$, $X_n = U$ and $X = 1 - U$, as in the exercise of Part II: $X_n \xrightarrow{d} X$, while $\|X_n - X\| = |2U - 1|$ does not tend to 0 in probability.

A constant limit upgrades the mode

Theorem. If $X_n \xrightarrow{d} c$ for a constant $c \in \mathbb{R}^d$ then $X_n \xrightarrow{\mathbb{P}} c$.

Proof. Let $\epsilon > 0$. The set $F = \{x : \|x - c\| \geq \epsilon\}$ is closed and does not contain c , so portmanteau (vi) gives

$$\limsup_n \mathbb{P}(\|X_n - c\| \geq \epsilon) = \limsup_n \mathbb{P}(X_n \in F) \leq \mathbb{P}(c \in F) = 0.$$

Combined with the previous theorem, for a constant limit the two modes coincide: $X_n \xrightarrow{\mathbb{P}} c \iff X_n \xrightarrow{d} c$. This is what makes the weak law of large numbers usable, and it is why Slutsky's hypothesis is stated with a constant.

Asymptotically equivalent sequences

Theorem. If $X_n \xrightarrow{d} X$ and $\|X_n - Y_n\| \xrightarrow{\mathbb{P}} 0$, then $Y_n \xrightarrow{d} X$.

Proof. Let f be bounded by M and L -Lipschitz, and $\epsilon > 0$. Then

$$|\mathbb{E}_{Y_n}[f(Y_n)] - \mathbb{E}_X[f(X)]| \leq \underbrace{\mathbb{E}_{X_n, Y_n}[|f(Y_n) - f(X_n)|]}_{\leq L\epsilon + 2M\mathbb{P}(\|X_n - Y_n\| \geq \epsilon)} + \underbrace{|\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]|}_{\rightarrow 0 \text{ by portmanteau (iii)}},$$

so the limsup is at most $L\epsilon$ for every $\epsilon > 0$. Portmanteau (iii) again.

Reading. A perturbation that is negligible in probability is invisible in distribution. This is the workhorse of the whole course: every asymptotic normality proof ends by writing the estimator as a sum that satisfies a central limit theorem, plus a remainder that is $o_{\mathbb{P}}(1)$.

Combining two convergences

Theorem. Let (X_n) , (Y_n) , X , Y be random vectors and c a constant vector.

1. **(Slutsky)** If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{\mathbb{P}} c$ then $(X_n, Y_n) \xrightarrow{d} (X, c)$.
2. If $X_n \xrightarrow{\mathbb{P}} X$ and $Y_n \xrightarrow{\mathbb{P}} Y$ then $(X_n, Y_n) \xrightarrow{\mathbb{P}} (X, Y)$.
3. If $X_n \xrightarrow{a.s.} X$ and $Y_n \xrightarrow{a.s.} Y$ then $(X_n, Y_n) \xrightarrow{a.s.} (X, Y)$.

Proof of the combination theorem

Item 1. Since $Y_n \xrightarrow{\mathbb{P}} c$, we have $\|(X_n, Y_n) - (X_n, c)\| = \|Y_n - c\| \xrightarrow{\mathbb{P}} 0$, so by the previous theorem it is enough to show $(X_n, c) \xrightarrow{d} (X, c)$. For f bounded continuous on $\mathbb{R}^d \times \mathbb{R}^{d'}$, the section $f_c(x) = f(x, c)$ is bounded continuous, so

$$\mathbb{E}_{X_n}[f(X_n, c)] = \mathbb{E}_{X_n}[f_c(X_n)] \rightarrow \mathbb{E}_X[f_c(X)] = \mathbb{E}_X[f(X, c)].$$

Item 2. With the equivalent norm $\|(x, y)\| = \|x\| + \|y\|$, a union bound gives

$$\mathbb{P}(\|(X_n, Y_n) - (X, Y)\| \geq \epsilon) \leq \mathbb{P}(\|X_n - X\| \geq \frac{\epsilon}{2}) + \mathbb{P}(\|Y_n - Y\| \geq \frac{\epsilon}{2}) \rightarrow 0.$$

Item 3. The intersection of two events of probability 1 has probability 1, and on it both coordinates converge.

Exercise, marginals do not give the joint law

Claim. $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ do not imply $(X_n, Y_n) \xrightarrow{d} (X, Y)$.

Solution. Let X, Y be independent $\mathcal{N}(0, 1)$, and set $X_n = X$ and $Y_n = -X$ for every n . Then $Y_n \sim \mathcal{N}(0, 1) \stackrel{d}{=} Y$, so both marginal convergences hold. But $(X_n, Y_n) = (X, -X)$ is supported on the line $\{y = -x\}$, while (X, Y) has a density on $\mathbb{R}^2$, so the two laws differ and $(X_n, Y_n) \xrightarrow{d} (X, Y)$ fails.

What Slutsky repairs. When the second limit is a constant, the limit law on the product is forced to be $\mathbb{P}_X \otimes \delta_c$ whatever the dependence, because a degenerate coordinate cannot be correlated with anything.

The continuous mapping theorem

Theorem. Let $g : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be measurable, let C be the set of points at which g is continuous, and let X satisfy $\mathbb{P}(X \in C) = 1$. Then

$$1. X_n \xrightarrow{a.s.} X \implies g(X_n) \xrightarrow{a.s.} g(X).$$

$$2. X_n \xrightarrow{\mathbb{P}} X \implies g(X_n) \xrightarrow{\mathbb{P}} g(X).$$

$$3. X_n \xrightarrow{d} X \implies g(X_n) \xrightarrow{d} g(X).$$

Remark. If $X = c$ is a constant, the condition reduces to continuity of g at the single point c . That is the version used in practice, together with Slutsky.

Proof, the almost sure and the in probability cases

Item 1. On the event $\{X_n \rightarrow X\} \cap \{X \in C\}$, which has probability 1, continuity of g at the point $X(\omega)$ gives $g(X_n(\omega)) \rightarrow g(X(\omega))$.

Item 2. Let $\epsilon, \delta > 0$ and

$$B_\delta = \{x \in \mathbb{R}^d : \exists y, \|x - y\| \leq \delta \text{ and } \|g(x) - g(y)\| \geq \epsilon\}.$$

Splitting on $\{\|X_n - X\| \leq \delta\}$ and its complement,

$$\mathbb{P}(\|g(X_n) - g(X)\| \geq \epsilon) \leq \mathbb{P}(X \in B_\delta) + \mathbb{P}(\|X_n - X\| \geq \delta).$$

The second term tends to 0. For the first, every $x \in C$ leaves B_δ once δ is small enough, so $\mathbf{1}\{x \in B_\delta \cap C\} \rightarrow 0$ as $\delta \rightarrow 0$, and dominated convergence gives $\mathbb{P}(X \in B_\delta) = \mathbb{P}(X \in B_\delta \cap C) \rightarrow 0$. Let $n \rightarrow \infty$ first, then $\delta \rightarrow 0$.

Proof, the convergence in law case

Item 3. Let $F \subset \mathbb{R}^m$ be closed. Then $\{g(X_n) \in F\} = \{X_n \in g^{-1}(F)\}$, and

$$g^{-1}(F) \subset \overline{g^{-1}(F)} \subset g^{-1}(F) \cup C^c.$$

For the second inclusion, let $x \in \overline{g^{-1}(F)}$ and $x_k \rightarrow x$ with $g(x_k) \in F$. If $x \in C$ then $g(x_k) \rightarrow g(x)$, so $g(x) \in F$ since F is closed, hence $x \in g^{-1}(F)$. Otherwise $x \in C^c$.

Applying portmanteau (vi) to the closed set $\overline{g^{-1}(F)}$,

$$\limsup_n \mathbb{P}(g(X_n) \in F) \leq \limsup_n \mathbb{P}(X_n \in \overline{g^{-1}(F)}) \leq \mathbb{P}(X \in \overline{g^{-1}(F)}) \leq \mathbb{P}(X \in g^{-1}(F)) + \underbrace{\mathbb{P}(X \in C^c)}_{=0},$$

which is $\mathbb{P}(g(X) \in F)$. Portmanteau (vi) in the other direction gives $g(X_n) \xrightarrow{d} g(X)$.

Slutsky, the usable form

Corollary. If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$ with c constant, then

1. $X_n + Y_n \xrightarrow{d} X + c$, for $X_n, Y_n, c \in \mathbb{R}^d$,
2. $Y_n X_n \xrightarrow{d} cX$, for $X_n \in \mathbb{R}^d$ and $Y_n, c \in \mathbb{R}$,
3. $Y_n^{-1} X_n \xrightarrow{d} c^{-1} X$, for $X_n \in \mathbb{R}^d$ and $Y_n, c \in \mathbb{R} \setminus \{0\}$.

Proof. A constant limit in distribution is a limit in probability, so the combination theorem gives $(X_n, Y_n) \xrightarrow{d} (X, c)$. Apply continuous mapping with $g(x, y) = x + y$, then $g(x, y) = yx$, then $g(x, y) = x/y$. The first two are continuous everywhere, the third on $\{y \neq 0\}$, which carries the limit law since $c \neq 0$.

Uniform convergence of the distribution functions

Lemma (Polya). If $X_n \xrightarrow{d} X$ in $\mathbb{R}$ and F_X is continuous on $\mathbb{R}$, then the convergence is uniform:

$$\sup_{x \in \mathbb{R}} |F_{X_n}(x) - F_X(x)| \xrightarrow{n \rightarrow \infty} 0.$$

Proof. Fix an integer N . Since F_X is continuous with limits 0 and 1, the intermediate value theorem gives $x_1 < \cdots < x_{N-1}$ with $F_X(x_i) = i/N$; set $x_0 = -\infty, x_N = +\infty$. For $x \in [x_{i-1}, x_i]$, monotonicity of F_{X_n} and of F_X gives

$$F_{X_n}(x_{i-1}) - F_X(x_{i-1}) - \frac{1}{N} \leq F_{X_n}(x) - F_X(x) \leq F_{X_n}(x_i) - F_X(x_i) + \frac{1}{N},$$

using $F_X(x_i) - F_X(x_{i-1}) = 1/N$ on both sides.

Hence $\sup_x |F_{X_n}(x) - F_X(x)| \leq \max_{i \leq N} |F_{X_n}(x_i) - F_X(x_i)| + \frac{1}{N}$, a finite maximum, so $\limsup_n \sup_x |F_{X_n} - F_X| \leq 1/N$ for every N .

Convergence in law does not give convergence of moments

Let $\mathbb{P}(X_n = n) = 1/n$ and $\mathbb{P}(X_n = 0) = 1 - 1/n$. Then $X_n \xrightarrow{d} 0$, while $\mathbb{E}_{X_n}[X_n] = 1$ for every n . Portmanteau tests bounded functions only, and the identity is not bounded.

Definition. A sequence (X_n) is uniformly integrable if

$$\lim_{M \rightarrow \infty} \sup_{n \in \mathbb{N}} \mathbb{E}_{X_n}[\|X_n\| \mathbf{1}_{\{\|X_n\| \geq M\}}] = 0.$$

Theorem. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be continuous on a set C , let $X \in C$ almost surely, and let $X_n \xrightarrow{d} X$. If $(f(X_n))_n$ is uniformly integrable then $\mathbb{E}_{X_n}[f(X_n)] \rightarrow \mathbb{E}_X[f(X)]$.

Proof, and the usable criterion

Assume $f \geq 0$, the general case following by splitting into positive and negative parts. For every $M > 0$,

$$\begin{aligned} \limsup_n |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| &\leq \limsup_n |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_{X_n}[f(X_n) \wedge M]| \\ &\quad + \limsup_n |\mathbb{E}_{X_n}[f(X_n) \wedge M] - \mathbb{E}_X[f(X) \wedge M]| \\ &\quad + |\mathbb{E}_X[f(X) \wedge M] - \mathbb{E}_X[f(X)]|. \end{aligned}$$

The first line is at most $\sup_n \mathbb{E}_{X_n}[f(X_n)\mathbf{1}\{f(X_n) \geq M\}]$, small for M large by uniform integrability. The third is small for M large by monotone convergence. The middle one is 0 for every M , since $f \wedge M$ is bounded and continuous on C , so continuous mapping and portmanteau (ii) apply.

Criterion. If $\sup_n \mathbb{E}_{X_n}[\|X_n\|^{1+\eta}] < \infty$ for some $\eta > 0$ then (X_n) is uniformly integrable, because $\mathbb{E}_{X_n}[\|X_n\|\mathbf{1}\{\|X_n\| \geq M\}] \leq M^{-\eta}\mathbb{E}_{X_n}[\|X_n\|^{1+\eta}]$. A uniformly bounded moment of order strictly above 1 is what one checks in practice.

Part VI. A first example in statistics

A proportion

Let $X_1, \dots, X_n$ be i.i.d. $\text{Ber}(p)$ with $p \in (0, 1)$ unknown, and $\bar{X}_n = \frac{1}{n} \sum_{i \leq n} X_i$. The question is what can be said about p from the sample, with no assumption beyond the model.

Law of large numbers. $\mathbb{E}_X[X] = p$ and $X \in L^1$, so $\bar{X}_n \xrightarrow{\mathbb{P}} p$. The estimator is consistent, which says nothing about its error.

Central limit theorem. $\text{Var}_X(X) = p(1 - p) < \infty$, so

$$\sqrt{n}(\bar{X}_n - p) \xrightarrow{d} \mathcal{N}(0, p(1 - p)).$$

The error is of order $n^{-1/2}$ and its limiting law is known. It is not yet usable: the scale $p(1 - p)$ involves the unknown.

Studentisation

Exercise. Show that

$$\frac{\sqrt{n}(\bar{X}_n - p)}{\sqrt{\bar{X}_n(1 - \bar{X}_n)}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Solution. From $\bar{X}_n \xrightarrow{\mathbb{P}} p$ and continuous mapping with $u \mapsto u(1 - u)$, continuous on $\mathbb{R}$,

$$\bar{X}_n(1 - \bar{X}_n) \xrightarrow{\mathbb{P}} p(1 - p).$$

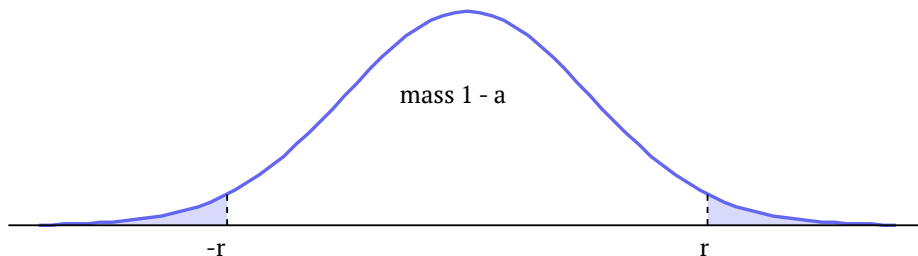
By Slutsky, the pair converges jointly,

$$(\sqrt{n}(\bar{X}_n - p), \bar{X}_n(1 - \bar{X}_n)) \xrightarrow{d} (\mathcal{N}(0, p(1 - p)), p(1 - p)),$$

and continuous mapping with $g(u, v) = u/\sqrt{v}$ finishes, since g is continuous on $\{v > 0\}$ and the limit lives there almost surely because $p \in (0, 1)$. The limit is $\mathcal{N}(0, p(1 - p))/\sqrt{p(1 - p)} \stackrel{d}{=} \mathcal{N}(0, 1)$.

An asymptotic confidence interval

Let r_α be the quantile of order $1 - \alpha/2$ of $\mathcal{N}(0, 1)$, so that $\mathbb{P}(|Z| \leq r_\alpha) = 1 - \alpha$ for $Z \sim \mathcal{N}(0, 1)$.



By the previous slide and portmanteau (vii) applied to $B = [-r_\alpha, r_\alpha]$, whose boundary has zero mass under $\mathcal{N}(0, 1)$,

$$\mathbb{P} \left(\frac{\sqrt{n}(\bar{X}_n - p)}{\sqrt{\bar{X}_n(1 - \bar{X}_n)}} \in [-r_\alpha, r_\alpha] \right) \xrightarrow{n \rightarrow \infty} 1 - \alpha.$$

Reading the interval

Inverting the event, the random interval

$$I_n = \left[\bar{X}_n - r_\alpha \sqrt{\frac{\bar{X}_n(1 - \bar{X}_n)}{n}}, \bar{X}_n + r_\alpha \sqrt{\frac{\bar{X}_n(1 - \bar{X}_n)}{n}} \right] \text{ satisfies } \mathbb{P}(p \in I_n) \rightarrow 1 - \alpha.$$

It is computable. Every quantity in I_n is a function of the sample and of α . That is the entire purpose of the studentisation step, and it is why the plug-in estimate of the variance is used rather than $p(1 - p)$.

It is asymptotic. The statement is a limit, not an inequality at fixed n . For a guarantee at fixed n one needs a concentration inequality, which costs a worse constant and does not adapt to the variance.

It is not the only choice. Inverting instead $\{p : |\sqrt{n}(\bar{X}_n - p)| \leq r_\alpha \sqrt{p(1 - p)}\}$, a quadratic in p , gives the Wilson interval, asymptotically equivalent and much better for small n or p close to 0 or 1.

Part VII. The symbols $\mathcal{O}_{\mathbb{P}}$ and $\mathcal{O}_{\mathbb{P}}$

Two symbols

Definition. Let (X_n) be a sequence of random vectors and (R_n) a sequence of nonnegative random variables.

$X_n = o_{\mathbb{P}}(1)$ means $\|X_n\| \xrightarrow{\mathbb{P}} 0$. More generally $X_n = o_{\mathbb{P}}(R_n)$ means there is (Y_n) with $X_n = R_n Y_n$ for every n and $\|Y_n\| \xrightarrow{\mathbb{P}} 0$.

$X_n = O_{\mathbb{P}}(1)$ means (X_n) is uniformly tight. More generally $X_n = O_{\mathbb{P}}(R_n)$ means there is (Y_n) with $X_n = R_n Y_n$ for every n and (Y_n) uniformly tight.

The dictionary. $o_{\mathbb{P}}$ is the probabilistic version of "negligible" and $O_{\mathbb{P}}$ of "bounded", with convergence in probability replacing convergence to 0 and uniform tightness replacing boundedness. Prohorov item 1 says that $X_n \xrightarrow{d} X$ implies $X_n = O_{\mathbb{P}}(1)$, which is the source of almost every $O_{\mathbb{P}}$ in the course.

Transferring o and O to $o_{\mathbb{P}}$ and $O_{\mathbb{P}}$

Theorem. Let $X_n \xrightarrow{\mathbb{P}} 0$ in $\mathbb{R}^d$, let $q > 0$ and let $R : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be measurable with $R(0) = 0$. Then

1. $\|R(h)\| = o(\|h\|^q)$ as $h \rightarrow 0$ implies $R(X_n) = o_{\mathbb{P}}(\|X_n\|^q)$,
2. $\|R(h)\| = O(\|h\|^q)$ as $h \rightarrow 0$ implies $R(X_n) = O_{\mathbb{P}}(\|X_n\|^q)$.

Why it is needed. A Taylor expansion produces a deterministic remainder bound, valid as the increment goes to 0. Every asymptotic expansion of an estimator evaluates that remainder at a random increment, and this theorem is what licenses the substitution.

Proof of the transfer

Define $g : \mathbb{R}^d \rightarrow \mathbb{R}^m$ by $g(h) = R(h)/\|h\|^q$ for $h \neq 0$ and $g(0) = 0$, so that $R(X_n) = g(X_n)\|X_n\|^q$ identically.

Item 1. The hypothesis $\|R(h)\| = o(\|h\|^q)$ says exactly $g(h) \rightarrow 0 = g(0)$ as $h \rightarrow 0$, so g is continuous at 0. Since $X_n \xrightarrow{\mathbb{P}} 0$, continuous mapping gives $g(X_n) \xrightarrow{\mathbb{P}} 0$, which is the claim.

Item 2. There are $M, \delta > 0$ with $\|R(h)\| \leq M\|h\|^q$ for $\|h\| \leq \delta$, hence $\|g(h)\| \leq M$ on that ball. Then

$$\limsup_n \mathbb{P}(\|g(X_n)\| \geq M) \leq \limsup_n \mathbb{P}(\|X_n\| \geq \delta) = 0,$$

so $(g(X_n))$ is uniformly tight, and $R(X_n) = O_{\mathbb{P}}(\|X_n\|^q)$.

Exercise, $o_{\mathbb{P}}(O_{\mathbb{P}}(1)) = o_{\mathbb{P}}(1)$

Claim. If $X_n = O_{\mathbb{P}}(1)$ and $Y_n = o_{\mathbb{P}}(X_n)$, then $Y_n = o_{\mathbb{P}}(1)$.

Solution. By definition there is (M_n) with $Y_n = M_n X_n$ and $M_n \xrightarrow{\mathbb{P}} 0$. Fix $\epsilon > 0$ and split on the size of X_n : for any $M > 0$,

$$\mathbb{P}(|Y_n| > \epsilon) = \mathbb{P}(|M_n X_n| > \epsilon, |X_n| \geq M) + \mathbb{P}(|M_n X_n| > \epsilon, |X_n| < M) \leq \mathbb{P}(|X_n| \geq M) + \mathbb{P}(|M_n| > \epsilon/M)$$

Given $\eta > 0$, choose M with $\sup_n \mathbb{P}(|X_n| \geq M) \leq \eta$, which is possible because (X_n) is uniformly tight.

With M now fixed, the second term tends to 0 since $M_n \xrightarrow{\mathbb{P}} 0$. Hence $\limsup_n \mathbb{P}(|Y_n| > \epsilon) \leq \eta$ for every $\eta > 0$.

The rules of calculation

The same two step split proves the rest of the calculus. For sequences of random vectors:

$o_{\mathbb{P}}(1) + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1)$ and $O_{\mathbb{P}}(1) + O_{\mathbb{P}}(1) = O_{\mathbb{P}}(1)$, by a union bound on the two halves of ϵ , respectively of M .

$o_{\mathbb{P}}(1) + O_{\mathbb{P}}(1) = O_{\mathbb{P}}(1)$ and $o_{\mathbb{P}}(1)O_{\mathbb{P}}(1) = o_{\mathbb{P}}(1)$, the latter being the exercise above.

$X_n \xrightarrow{d} X \implies X_n = O_{\mathbb{P}}(1)$, which is Prohorov item 1, and $X_n = O_{\mathbb{P}}(a_n)$ with $a_n \rightarrow 0$ implies $X_n = o_{\mathbb{P}}(1)$.

A warning. These are one way statements. From $X_n = O_{\mathbb{P}}(1)$ nothing follows about convergence of any kind, tightness being compactness and not convergence.